# AI.405

## Robot Manipulation

### COURSE OVERVIEW

Robot grasping and manipulation: grippers, grasp planning, dexterous manipulation, and contact.

### TACT HOURS

dule

ROBOT MANIPULATION

1

SESSION

60 MINUTES

# ORIENTATION: GRASPING THE WORLD

**KEY TOPICS**

- Manipulation in robotics
- Grasp types
- Contact mechanics
- Applications

**TEACHING FOCUS**

Let robots hold and move objects.

**KEY TAKEAWAY** Manipulation is physical intelligence.

2

SESSION

60 MINUTES

# GRIPPERS & END-EFFECTORS

**KEY TOPICS**

- Parallel-jaw grippers
- Vacuum and adhesive
- Multi-fingered hands
- End-effector selection

**TEACHING FOCUS**

Choose the right hand for the task.

▲ **KEY TAKEAWAY** The gripper is the robot's hand.

ROBOT MANIPULATION

3

SESSION

60 MINUTES

## GRASP PLANNING

**KEY TOPICS**

- Grasp taxonomy
- Force-closure and form-closure
- Grasp quality metrics
- Grasp synthesis

**TEACHING FOCUS**

Plan stable and effective grasps.

**KEY TAKEAWAY** A good grasp resists disturbance.

4

SESSION

60 MINUTES

## DEXTEROUS MANIPULATION

**KEY TOPICS**

- In-hand manipulation
- Finger gaiting
- Tactile feedback
- Dexterous control

**TEACHING FOCUS**

Manipulate objects skillfully.

**KEY TAKEAWAY** Dexterity turns grasping into tool use.

5

SESSION

60 MINUTES

## CONTACT & INTERACTION

**KEY TOPICS**

- Contact modeling
- Impedance control
- Slip detection
- Safe physical interaction

**TEACHING FOCUS**

Manage contact safely.

**KEY TAKEAWAY** Contact is where manipulation happens.

ROBOT MANIPULATION

6

SESSION

60 MINUTES

## ROBOT MANIPULATION & REVIEW

**KEY TOPICS**

- Manipulation demo
- Course recap and assessment
- Toward robot perception (AI.406)
- Grasp challenge

**TEACHING FOCUS**

Pick, place, and manipulate reliably.

**KEY TAKEAWAY** Manipulation is the robot's handiwork.